

Advances for Digital Humans in VFX Production at Goodbye Kansas Studios

Rasmus Haapaoja
rasmus.haapaoja@goodbyekansas.com
Goodbye Kansas Studios
Stockholm, Sweden

Josefine Klintberg
josefine.klintberg@goodbyekansas.com
Goodbye Kansas Studios
Stockholm, Sweden



Figure 1: A digital human created and rendered at Goodbye Kansas Studios.

ABSTRACT

Digital humans and digital doubles are core products at Goodbye Kansas Studios and in order to further improve their quality and related workflows, a number of new tools have been developed and integrated into our existing pipeline.

This talk will cover some of these implementations, such as an offline generation of blendshapes based on facial scans, our take on the sticky lips problem and an efficient implementation of render time dynamic skin microstructure. Furthermore, we look into future usages and how *Universal Scene Description* (USD) [Pixar Animation Studios 2021] can serve as a helpful tool for facial rigs.

CCS CONCEPTS

• Computing methodologies → Computer graphics.

KEYWORDS

facial rigging, blendshapes, sticky lips, skin microstructure, displacement

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1 BLENDSHAPES FROM FACIAL SCANS

Using our advanced in-house 3D scanning and capture studio, we can obtain highly detailed input for our face department to develop facial rigs from. Currently, the transferring of details from scans to rig require extensive manual work, and is an area where we wished to automate the workflow.

Aiming for an offline approach where the generated blendshapes are not averaged between scans, the method by [Seol et al. 2016] serves as a basis for our implementation, where the transfer is set from a single scan. We tweak the approach to work solely on subsets of the rig, chosen by an artist, to use displacement vectors in the initial transfer and perform alignment using least-squares fitting and vertex masks. We also extend the method to target corrective blendshapes.

With corrective blendshapes included in the transfer process, we found that it is crucial to express these as relative shapes – not absolute – but still let them be driven from the generic rig. The tool outputs valid and updated corrective shapes, although incorrect vertex distributions can be seen when using many overlapping shapes. Despite this limitation, we provide a flexible and efficient tool with a quick processing time and an alterable configuration.

1.1 Integrating USD

The adaptation and integration of USD at the studio provided interesting viewpoints on how to make it easier to inspect and iterate on facial rigs in the future. Our new facial tools support USD files as input where the rig is represented as a single USD asset, with individual blendshapes referenced as sublayers. With the quick 3D overview in the usdviewer, an easy switch between variants and versions and the possibility to add parameters to specific blendshapes, USD shows many promises for future efficient handling of facial rigs.

2 STICKY LIPS

The fine details of the mouth and how it behaves when moving is a key aspect to the perceived realism of a digital human, with the subtle effect of stickiness between the lips being an important factor. Our implementation is based on a simplified spring model that allow us to avoid the manual work required by e.g. a curve based zip solution, as well as the time and stability concerns of a fully simulated approach.

First, we detect and solve collisions between the lips and attach springs between the contact points with a user specified stiffness, lifetime and stiffness falloff. The springs are detached from the collision points when their lifetime has ended and are not allowed to overshoot during simulation in order to guarantee stability.

We expose our sticky lips implementation to face rigging artists as a Maya deformer which operates in real-time and requires minimal setup, apart from painting a mask defining where the lips are located. A comparison with and without our sticky lips deformer on an animated character can be found in Figure 2.



Figure 2: Sticky lips on (left) and off (right).

3 DYNAMIC SKIN MICROSTRUCTURE

The skin of a human is highly dynamic and facial expressions deform the skin microstructure in different ways. Our previous approach for replicating the microstructure, using a static high-resolution displacement map, failed to capture the subtle stretching and compression of features such as pores and tiny wrinkles. Our new approach, inspired by the method introduced by [Nagano et al. 2015], still uses a static displacement map as input but is dynamically filtered based on the deformation of the underlying skin mesh.

We start by establishing a mapping from the neutral to the deformed triangles through a per-triangle linear transform in tangent space, which is then averaged and stored as a per-vertex mesh attribute. The per-vertex transform is then interpolated in UV space, which allows us to extract the stretch, compression and orientations at each pixel. Two convolution kernels are further constructed, with size and orientation based on the per-pixel tension parameters. The dynamic displacement map is finally generated by applying these kernels to a static input map, performing blurring and sharpening in orthogonal directions.

The dynamic displacement map filtering is exposed to our look-development and lighting artists as a custom texture plugin for the V-Ray renderer in Houdini, allowing us to perform the filtering at render time. Furthermore, we utilize OpenCL kernels to perform the transform, interpolation and convolution passes, giving us the

ability to maximize the performance by making use of the GPU. Our custom texture node is not strictly limited to use the resulting displacement map. The internal buffers for stretching and compression across the mesh are also exposed, which may be used to drive other effects such as blood flow. A showcase of a displacement map which is filtered by our plugin and applied to a character can be seen in Figure 3.



Figure 3: Closeup of dynamic skin microstructure.

4 CONCLUSION

With an automatic transfer tool, we make it more efficient to create new facial rigs. Using simulations of observable physical properties such as sticky lips and skin microstructures, we can generate the fine details that are required when producing a realistic looking character. Gathering our new suite of production ready tools, together with technologies such as USD, we are building a strong toolbox in our pipeline that fits into our artists current workflows, allowing them to develop more detailed high-end characters than before.

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